

Skills Summary

Reading a Rational Graph

Use this reference while completing your worksheet or when studying.

Skill 1 — Simplify, and keep the domain

Rule: factor top and bottom completely, then cancel whole shared factors. The excluded values come from the *original* denominator — cancelling changes the formula, never the domain.

Example: Simplify $\frac{x^2 - 25}{x^2 - 3x - 10}$.

Step 1. Nothing can cancel while the parts are sums, so factor both of them before looking for a shared factor.

Step 2. $\frac{(x-5)(x+5)}{(x-5)(x+2)}$: the factor $(x-5)$ is shared, and the original denominator is zero at $x = 5$ and $x = -2$, so both stay excluded.

$$\Rightarrow \frac{x+5}{x+2}$$

Try it: Simplify $\frac{x^2 - 36}{x^2 + 2x - 24}$.

check: $\frac{4-x}{9-x}$

Skill 2 — Intercepts

Rule: an x -intercept is where the *simplified* numerator is zero (a zero that cancelled is a hole, not an intercept). The y -intercept is the value at $x = 0$.

Example: Find both intercepts of $f(x) = \frac{2x-6}{x+4}$.

Step 1. A quotient is zero only when its top is zero, and the y -intercept is just a substitution, so each intercept needs a different part of the formula.

Step 2. $2x - 6 = 0$ gives $x = 3$; and $f(0) = \frac{-6}{4}$.

$$x\text{-intercept } x = 3, \quad y\text{-intercept } -\frac{3}{2}$$

Try it: Find both intercepts of $g(x) = \frac{3x-12}{x+2}$.

check: $4-x$ and $9-x$

Skill 3 — Holes and vertical asymptotes

Rule: a denominator factor that *cancels* gives a hole, whose height is the simplified value there; a factor that *survives* gives a vertical asymptote at its zero.

Example: Locate the hole and the vertical asymptote of $f(x) = \frac{x^2 - 1}{x^2 - 3x + 2}$.

Step 1. Which zero is which is decided entirely by what cancels, so factor first and then sort the two denominator zeros.

Step 2. $\frac{(x-1)(x+1)}{(x-1)(x-2)} = \frac{x+1}{x-2}$: $(x-1)$ cancels, and at $x = 1$ the simplified form gives $\frac{2}{-1}$.
hole at $(1, -2)$; asymptote $x = 2$

Try it: Locate the hole and the vertical asymptote of $g(x) = \frac{x^2 - 4}{x^2 - x - 2}$.

check: hole at $(2, \frac{3}{4})$, asymptote $x = -1$

Skill 4 — Horizontal asymptotes from the degrees

Rule: compare the degree of the top with the degree of the bottom. Smaller on top gives $y = 0$; equal degrees give $y =$ the ratio of the *leading* coefficients; larger on top gives no horizontal asymptote. The constant terms never matter.

Example: Find the horizontal asymptote of $f(x) = \frac{5x^2 - 2x}{2x^2 + 7}$.

Step 1. Only the highest-power terms survive when x grows, so compare degrees first and read the leading coefficients.

Step 2. Both degrees are 2, so the level is $\frac{5}{2}$.

$$\Rightarrow y = \frac{5}{2}$$

Try it: Find the horizontal asymptote of $g(x) = \frac{3x + 8}{4x - 1}$.

check: $\frac{3}{4} = 0$